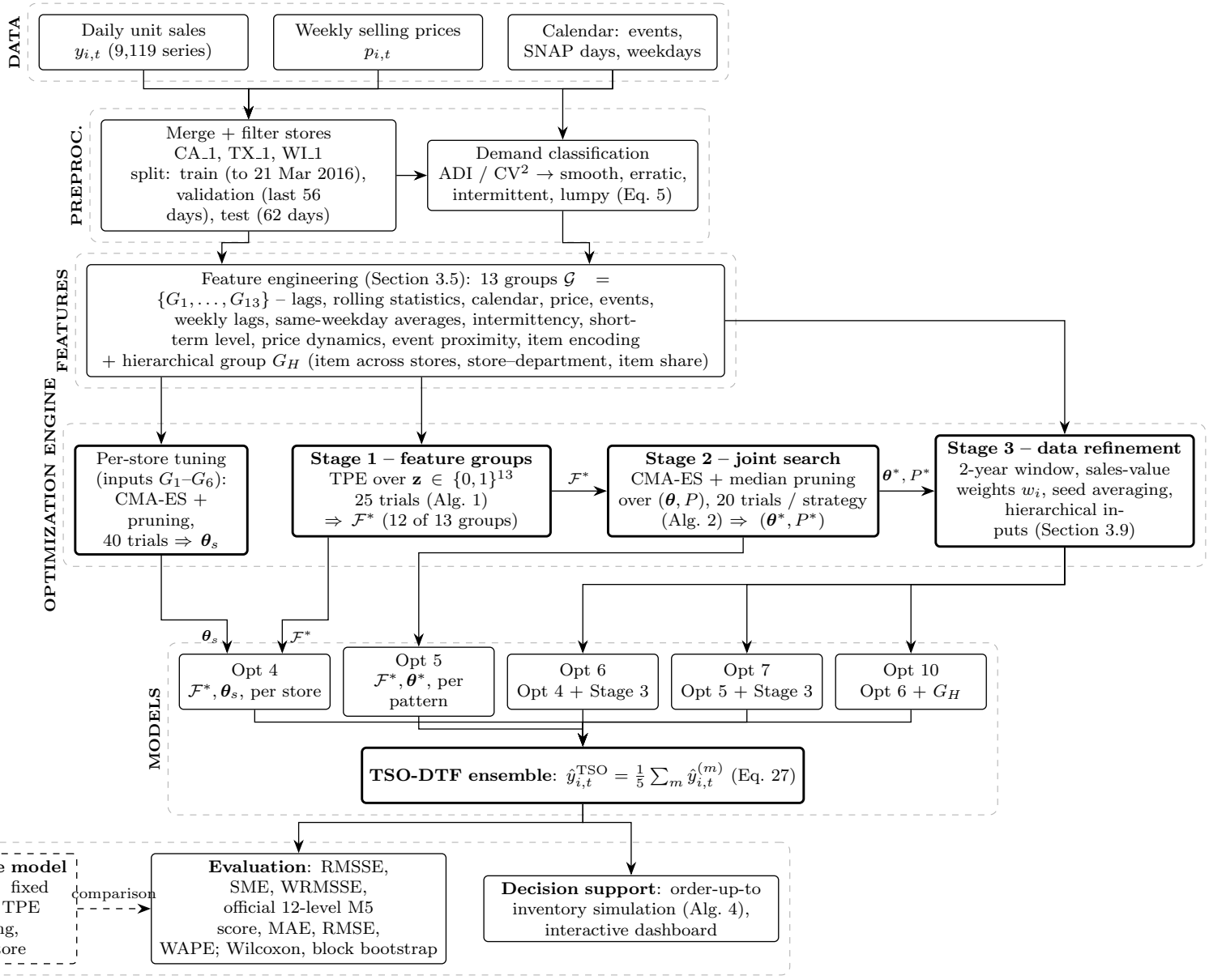

DATA
Daily unit sales $y_{i,t}$ (9,119 series)
Weekly selling prices $p_{i,t}$
Calendar: events, SNAP days, weekdays
PREPROC.
Merge + filter stores CA_1, TX_1, WI_1 split: train (to 21 Mar 2016), validation (last 56 days), test (62 days)
Demand classification ADI / CV$^2$ → smooth, erratic, intermittent, lumpy (Eq. 5)
FEATURES
Feature engineering (Section 3.5): 13 groups $\mathcal{G} = \{G_1, \ldots, G_{13}\}$ – lags, rolling statistics, calendar, price, events, weekly lags, same-weekday averages, intermittency, short-term level, price dynamics, event proximity, item encoding + hierarchical group $G_H$ (item across stores, store–department, item share)
OPTIMIZATION ENGINE
Per-store tuning (inputs $G_1$–$G_6$): CMA-ES + pruning, 40 trials ⇒ $\boldsymbol{\theta}_s$
Stage 1 – feature groups TPE over $\mathbf{z} \in \{0,1\}^{13}$ 25 trials (Alg. 1) ⇒ $\mathcal{F}^*$ (12 of 13 groups)
$\mathcal{F}^*$
Stage 2 – joint search CMA-ES + median pruning over $(\boldsymbol{\theta}, P)$, 20 trials / strategy (Alg. 2) ⇒ $(\boldsymbol{\theta}^*, P^*)$
$\boldsymbol{\theta}^*, P^*$
Stage 3 – data refinement 2-year window, sales-value weights $w_i$, seed averaging, hierarchical inputs (Section 3.9)
$\boldsymbol{\theta}_s$
$\mathcal{F}^*$
MODELS
Opt 4 $\mathcal{F}^*, \boldsymbol{\theta}_s$, per store
Opt 5 $\mathcal{F}^*, \boldsymbol{\theta}^*$, per pattern
Opt 6 Opt 4 + Stage 3
Opt 7 Opt 5 + Stage 3
Opt 10 Opt 6 + $G_H$
TSO-DTF ensemble: $\hat{y}^{\mathrm{TSO}}_{i,t} = \frac{1}{5}\sum_m \hat{y}^{(m)}_{i,t}$ (Eq. 27)
OUTPUT
Reference model DTF [6]: fixed inputs, TPE tuning, per store
comparison
Evaluation: RMSSE, SME, WRMSSE, official 12-level M5 score, MAE, RMSE, WAPE; Wilcoxon, block bootstrap
Decision support: order-up-to inventory simulation (Alg. 4), interactive dashboard